Week 6: Bernoulli's Equation

For steady, incompressible, inviscid flow along a streamline, Bernoulli's equation states

$$\frac{p}{\rho g} + \frac{v^2}{2g} + z = H$$

where p is the static pressure, ρ is the fluid density, v is the flow velocity, z is the elevation and H is the total head.

Hydrostatic pressure

The pressure at depth h below the free surface is

$$p = p_0 + \rho g h$$

where p_0 is the atmospheric pressure.

Continuity

For incompressible flow between two sections,

$$A_1 v_1 = A_2 v_2$$

Head loss

The Darcy–Weisbach equation gives the head loss as

$$h_f = f \frac{L}{D} \frac{v^2}{2g}$$

where f is the friction factor, L is the pipe length and D is the pipe diameter.

Reynolds number

$$Re = \frac{\rho v D}{\mu}$$

where μ is the dynamic viscosity.

Force on a vane

$$F_x = -\dot{m}(v_2 - v_1)$$